Software Requirements Specification (SRS) Template

Project: Online Quiz System
Version: 1.0
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Revision history

Version	Date	Author	Change summary	Approval
1.0	12-09-2026	Project Team	SRS for Online Quiz System	Student SRS

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for an Online Quiz System. It defines the functional and non-functional requirements, interfaces, and verification criteria for a console-based online MCQ quiz application.

1.2 Scope

The system allows students to register/login, access MCQ quizzes from an online server, answer questions, submit the quiz, receive a score, and view results. An administrator can manage quiz questions and view results. The system is online based and not a local-only quiz.

1.3 Audience

Students, project developers, testers, instructors, and project evaluators.

1.4 Definitions

List of acronyms: MCQ, SRS, FR, NFR, UI, API, HTTP, HTTPS.

2. Overall description

2.1 Product perspective

The Online Quiz System is a client-server application. The student uses a C++ console program that connects to an online server to obtain quizzes and store results.

2.2 Major product functions (detailed)

- User registration and login
- View and select available quizzes
- Retrieve MCQs and options online
- Answer and submit quiz
- Calculate and display score
- Store and view results online
- Display leaderboard
- Administrator question and result management

2.3 User roles and characteristics (expanded)

- Student: takes online quizzes and views scores.
- Administrator: manages questions and views results.
- Online Server: provides quiz data and stores results.

2.4 Operating environment

C++ console application on Windows or Linux with Internet connectivity. An online server and data store are used for quiz and result information.

2.5 Constraints

The application shall be console based. C++ shall be used. Internet connectivity is required for normal operation.

3. External interface requirements

3.1 User interfaces

The system shall provide a simple text-based console interface with numbered menus, clear prompts, MCQ options, score display, and error messages.

3.2 Hardware interfaces

No special hardware is required. A computer with keyboard, display, and network connectivity is sufficient.

3.3 Software interfaces

- Online Quiz Server/API for login, quiz retrieval, submission, and results.
- Online database/data store for users, questions, quizzes, and scores.
- C++ networking library for client-server communication.

3.4 Communications

The C++ client shall communicate with the online server using request/response messages. HTTPS should be used for secure online communication.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow OQS-F-###.

4.1 Student Quiz Functions

The system shall provide the following functional requirements.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQS-F-001	The system shall allow a student to register with a unique username and password.	Functional	High	Student	Valid details create an account. TC-F-01	Online server
OQS-F-002	The system shall authenticate a registered student	Functional	High	Student	Valid login opens student menu. TC-F-02	Online server

	using username and password.					
OQS-F-003	The system shall display available quizzes obtained from the online server.	Functional	High	Student	Available quizzes are displayed. TC-F-03	Internet required
OQS-F-004	The system shall allow a student to select an available quiz.	Functional	High	Student	Selected quiz is opened. TC-F-04	Active quiz
OQS-F-005	The system shall retrieve the selected quiz questions and options from the online server.	Functional	High	Student	Questions and options are displayed. TC-F-05	Server/API
OQS-F-006	The system shall display one MCQ at a time with its answer options.	Functional	High	Student	Question and options are clear. TC-F-06	Console UI
OQS-F-007	The system shall accept and validate the student's answer option.	Functional	High	Student	Only valid options are accepted. TC-F-07	Input validation
OQS-F-008	The system shall allow the student to move through all questions in the selected quiz.	Functional	High	Student	Student can answer each question. TC-F-08	Quiz data

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQS-F-009	The system shall allow the student to submit the completed quiz.	Functional	High	Student	Quiz is submitted after confirmation. TC-F-09	Online connection
OQS-F-010	The system shall calculate the score by comparing submitted answers with correct answers.	Functional	High	Student	Expected score is calculated. TC-F-10	Scoring
OQS-F-011	The system shall display the final score, correct answers, total questions, and percentage.	Functional	High	Student	Result summary is displayed. TC-F-11	After submission
OQS-F-012	The system shall store the student's quiz result on the online server.	Functional	High	Student	Server confirms result storage. TC-F-12	Online database
OQS-F-013	The system shall allow a student to view previous quiz results.	Functional	Medium	Student	Stored attempts are displayed. TC-F-13	Online database
OQS-F-014	The system shall display a leaderboard for quizzes where	Functional	Medium	Student	Leaderboard is displayed. TC-F-14	Online results

	ranking is enabled.					
OQS-F-015	The system shall allow an administrator to add, edit, or delete quiz questions and view student results.	Functional	High	Administrator	Authorized action succeeds. TC-F-15	Admin access

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow OQS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
OQS-NF-001	Normal quiz and result requests should receive a server response within 3 seconds.	Performance	High	90% meet 3 seconds. TC-NF-01
OQS-NF-002	The console shall provide clear menus, prompts, and error messages.	Usability	High	Usability test passes. TC-NF-02
OQS-NF-003	The online service shall target at least 99% monthly availability excluding planned maintenance.	Reliability	Medium	Availability ≥99%. TC-NF-03
OQS-NF-004	The system shall support at least 50 concurrent student sessions in the intended project setup.	Scalability	Medium	50-session test passes. TC-NF-04
OQS-NF-005	Important events such as login attempts, quiz submissions, and admin changes shall be timestamped in logs.	Auditability	High	Required events are logged. TC-NF-05

5.1. Security

5.1.1 Security Objectives

1. Protect user credentials, quiz data, and result data from unauthorized access.
2. Prevent unauthorized changes to quiz content and stored scores.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
OQS-SR-001	The system shall use HTTPS/TLS for communication between the client and online server.	Security	High	Encrypted communication verified. TC-SEC-01
OQS-SR-002	The system shall store passwords using secure hashing and shall not store plaintext passwords.	Security	High	Password storage check passes. TC-SEC-02
OQS-SR-003	The system shall restrict administrator functions to authenticated administrator accounts.	Security	High	Student access is denied. TC-SEC-03
OQS-SR-004	The server shall validate that a quiz submission belongs to the authenticated student before storing it.	Security	High	Unauthorized submission rejected. TC-SEC-04
OQS-SR-005	The system shall record failed logins and administrator actions without recording passwords.	Security	High	Audit log contains events, no password. TC-SEC-05

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: all functional requirements implemented and verified, with no critical NFR or security failures.
- Acceptance test suites: Login, Quiz, Submission, Scoring, Performance, Security, and Usability tests.

7. System models and diagrams

At least 2 UML use case diagrams are included for the Online Quiz System.

Figure 1: Student and Online Server Use-Case Diagram

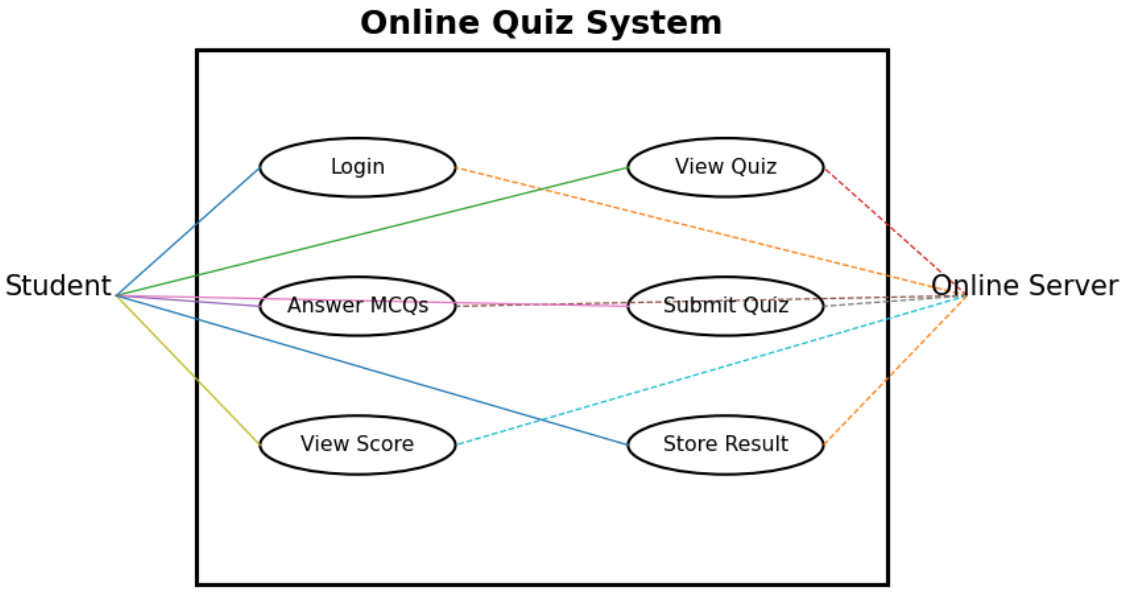
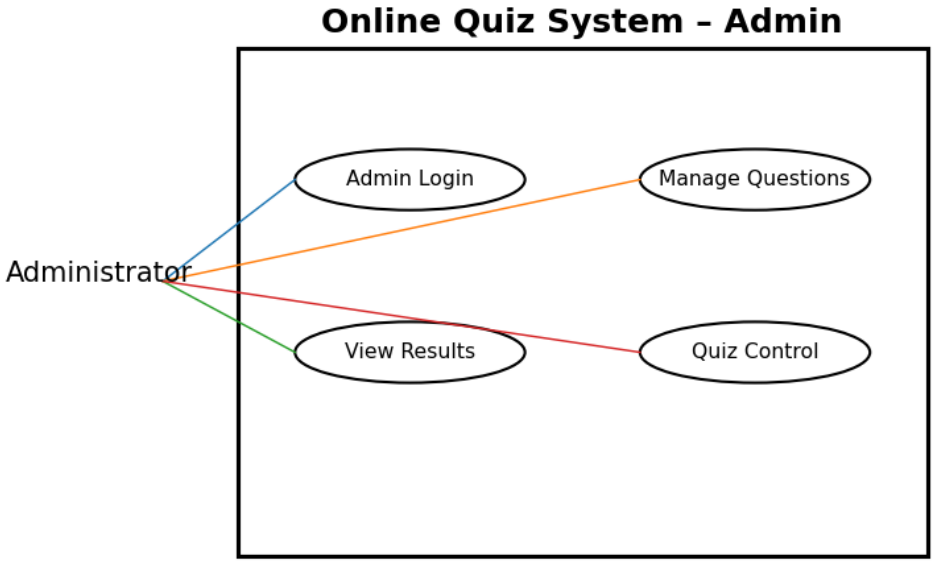


Figure 2: Administrator Use-Case Diagram



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
OQS-F-001	Register user	4 / DS-Auth	Auth Module	TC-F-01	N	
OQS-F-002	Login	4 / DS-Auth	Auth Module	TC-F-02	N	
OQS-F-003	View quizzes	4 / DS-Quiz	Quiz Module	TC-F-03	N	
OQS-F-005	Fetch MCQs	4 / DS-Quiz	Quiz Module	TC-F-05	N	
OQS-F-009	Submit quiz	4 / DS-Submit	Quiz Module	TC-F-09	N	
OQS-F-010	Calculate score	4 / DS-Score	Scoring Module	TC-F-10	N	
OQS-F-012	Store result	4 / DS-Result	Result Module	TC-F-12	N	
OQS-F-015	Admin functions	4 / DS-Admin	Admin Module	TC-F-15	N	
OQS-NF-001	Response time	5 / DS-Perf	Online Server	TC-NF-01	N	
OQS-SR-001	HTTPS/TLS	5.1.2 / DS-Sec	Network Layer	TC-SEC-01	N	